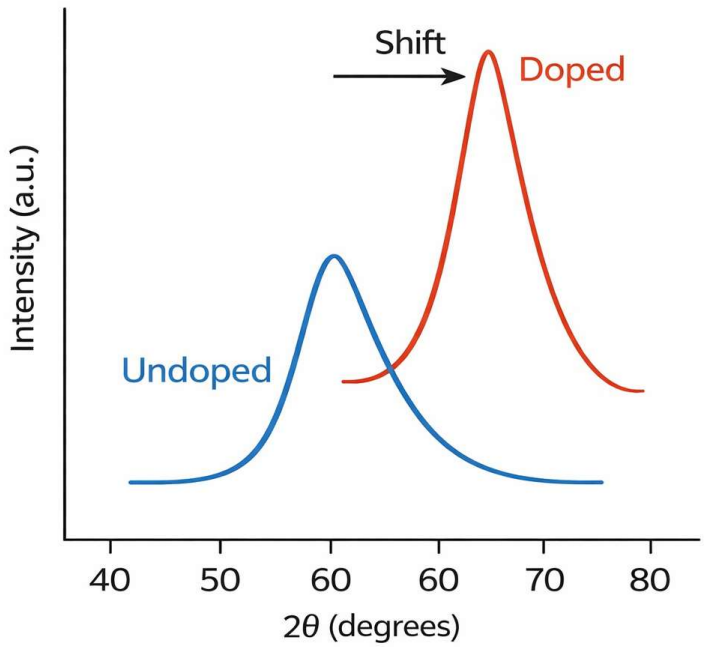
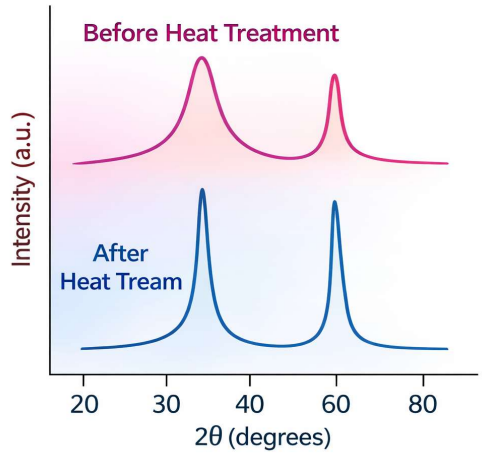
NANO LAB QUESTION BANK (A)

1. **Compare the XRD patterns of pure and doped nanoparticles shown below and explain the reason for the observed peak shift.**



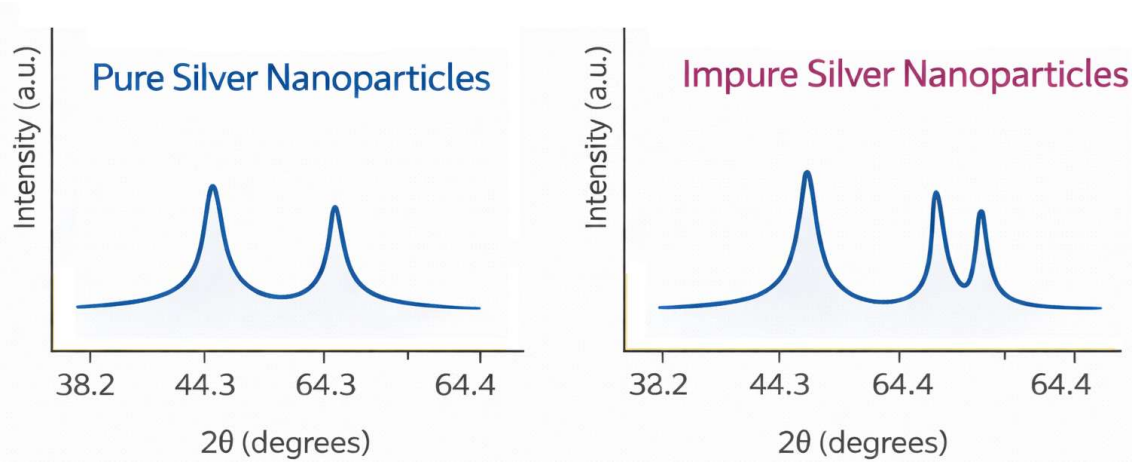
Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Peak Position (2θ)	Peak shifts to **higher 2θ** in doped sample	According to Bragg's Law, increase in θ occurs when **d-spacing decreases**	Indicates **lattice contraction**
d-spacing	d-spacing **decreases** after doping	Smaller dopant ions replace larger host atoms → atomic planes move closer	Confirms **reduction in interplanar spacing**
Lattice Structure	No new peaks; structure remains similar	Dopant ions enter lattice substitutionally without forming new phase	Confirms **successful doping without phase change**
Lattice Strain	Compressive strain develops	Size mismatch between dopant and host atoms creates internal stress	Causes **shift toward higher 2θ**
Peak Shape/Width *(optional)*	Slight change or narrowing may occur	Reduction in defects or improved crystallinity after doping/processing	Indicates **better structural ordering**
Overall Interpretation	System shows peak shift but same phase	Atomic-level substitution affects lattice parameters only	Confirms **dopant incorporation at atomic level**

1. **Two XRD patterns of a material (before and after heat treatment) are provided. After heat treatment, the peaks become sharper and more intense. Interpret the structural changes.**



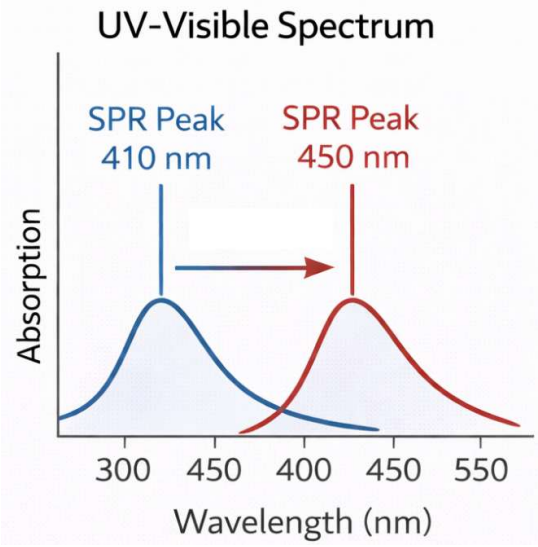
Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Peak Shape (FWHM)	Peaks become **sharper (narrower)** after heat treatment	Reduction in peak broadening due to **increase in crystallite size** and decrease in microstrain (Scherrer relation: $D \propto 1/\beta$)	Indicates **grain growth and improved crystallinity**
Peak Intensity	Peaks become **more intense**	Better atomic ordering leads to **more constructive diffraction** of X-rays	Confirms **increase in crystallinity**
Crystallinity	Higher crystallinity after heat treatment	Heat provides energy for atoms to rearrange into a **more ordered crystal structure**	Material becomes **more crystalline**
Crystallite Size	Crystallite size **increases**	Small crystallites merge during annealing, reducing grain boundaries	Indicates **grain growth**
Lattice Defects	Reduction in defects (dislocations, strain)	Heat treatment enables **atomic diffusion**, repairing defects	Results in **lower microstrain and improved structure**
Peak Position (2θ)	No significant shift in peak position (generally)	Lattice parameters remain unchanged (no doping or substitution)	Confirms **same phase and structure retained**
Overall Structure	More ordered and stable structure	Combined effect of grain growth and defect reduction	Material shows **enhanced structural stability**

2. **An XRD pattern shows an additional peak that does not match the expected phase. What are the possible interpretations?**



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Number of Peaks	Pure sample shows **2 peaks**, impure sample shows **3 peaks**	Additional crystalline phase contributes extra diffraction peak	Indicates **presence of impurity or secondary phase**
Peak Positions	Common peaks appear at similar 2θ positions in both samples	Main peaks correspond to the same silver crystal planes	Confirms **base material is still silver (Ag)**
Additional Peak	Extra peak appears only in impure sample	Arises from **impurity, unreacted precursor, or secondary phase**	Confirms **sample is not phase pure**
Phase Purity	Pure sample shows fewer peaks; impure shows more peaks	Each phase produces its own characteristic diffraction peaks	Pure = **single phase**, Impure = **multi-phase**
Pattern Complexity	Simple pattern (pure) vs complex pattern (impure)	More phases → more diffraction signals	Indicates **structural heterogeneity in impure sample**
Overall Structure	Pure sample is uniform; impure has mixed structure	Coexistence of different crystal structures	Confirms **impurity presence and incomplete purity**

3. **A UV–Visible spectrum of silver nanoparticles shows that the Surface Plasmon Resonance (SPR) peak shifts from 410 nm to 450 nm after 24 hours of storage. Interpret the significance of this red shift in terms of changes in nanoparticle size, aggregation behavior, and the refractive index of the surrounding medium.**

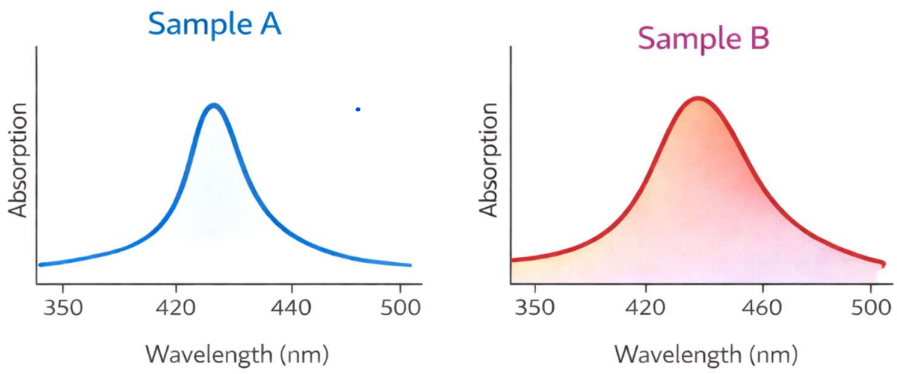


Aspect	Observation	Reason (Scientific Explanation)	Conclusion
SPR Peak Position	SPR peak shifts from **410 nm → 450 nm (red shift)**	Increase in wavelength indicates lower energy oscillation of surface electrons	Indicates **change in nanoparticle environment or structure**
Particle Size	Effective particle size appears to **increase**	Larger particles have stronger electron oscillations → absorb at longer wavelengths	Suggests **growth or increase in particle size**
Aggregation Behavior	Possible **aggregation of nanoparticles**	Close proximity leads to **plasmon coupling**, shifting SPR to longer wavelength	Confirms **particle clustering or agglomeration**
Peak Broadening (if observed)	SPR peak may become broader	Size distribution increases due to aggregation	Indicates **polydispersity and instability**
Refractive Index of Medium	Surrounding medium effect increases	Higher refractive index slows electron oscillation → red shift	Suggests **change in medium or adsorption on surface**
Stability Over Time	Change observed after 24 hours	Surface ligands/capping agents weaken → particles interact	Indicates **reduced stability of nanoparticles**
Overall Interpretation	Red shift with possible broadening	Combined effect of size increase, aggregation, and medium change	Confirms **aging-induced instability in nanoparticle system**

4. **A UV–Visible spectrum of nanoparticles shows two different patterns:**

- **Sample A exhibits a sharp and narrow SPR peak around 420 nm.**
- **Sample B shows a broad and red-shifted SPR peak around 460 nm.**

Based on the given data, differentiate between monodispersed and aggregated nanoparticles. Explain the reasons for peak broadening and red shift in Sample B.

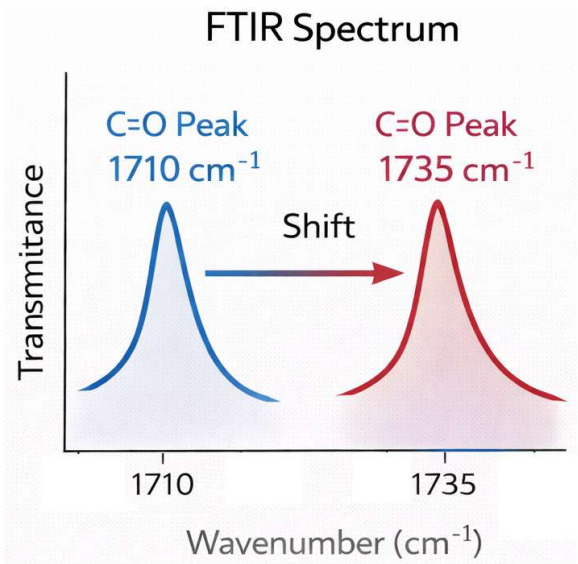


UV-Visible Spectra: Monodispersed vs Aggregated Nanoparticles

Aspect	Observation	Reason (Scientific Explanation)	Conclusion
SPR Peak Position	Sample A: Peak at ~420 nm; Sample B: Peak shifts to ~460 nm	Red shift occurs due to plasmon coupling when particles come closer	Sample B shows aggregation
Peak Shape	Sample A: Sharp and narrow peak; Sample B: Broad peak	Uniform size → narrow peak; Aggregation → wide size distribution → broadening	Sample A is monodispersed, Sample B is polydispersed
Particle Distribution	Sample A: Uniform particles; Sample B: Clusters of particles	Aggregation leads to interaction between nanoparticles	Sample B has particle clustering
Optical Behavior	Sample A: Single well-defined SPR band; Sample B: Broad and shifted SPR band	Inter-particle interaction modifies electron oscillation	Confirms plasmonic coupling in Sample B

Stability	Sample A: Stable system; Sample B: Less stable system	Weakening of capping agents causes aggregation	Sample A is stable, Sample B is unstable
Size Interpretation	Sample A: Smaller effective size; Sample B: Larger effective size (clusters)	Aggregates behave like larger particles optically	Sample B shows increased effective particle size
Overall Interpretation	Clear difference in spectra	Combined effect of size distribution and aggregation	Sample A = monodispersed, Sample B = aggregated nanoparticles

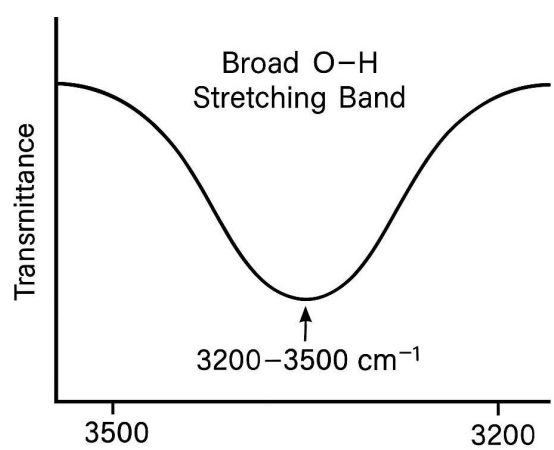
5. An FTIR spectrum of a polymer shows the C=O stretching peak shifting from 1710 cm^{-1} to 1735 cm^{-1} after chemical modification. Explain the structural or environmental factors that could cause this shit.



Aspect	**Observation**	**Reason (Scientific Explanation)**	**Conclusion**
Peak Position (C=O stretch)	Peak shifts from 1710 cm^{-1} → 1735 cm^{-1}	Increase in wavenumber indicates stronger C=O bond (higher bond energy)	Suggests change in chemical environment of carbonyl group
Electron-Withdrawing Groups (EWG)	Peak shifts to higher frequency	EWGs reduce electron density on C=O → increase bond strength →	Indicates introduction of electron-

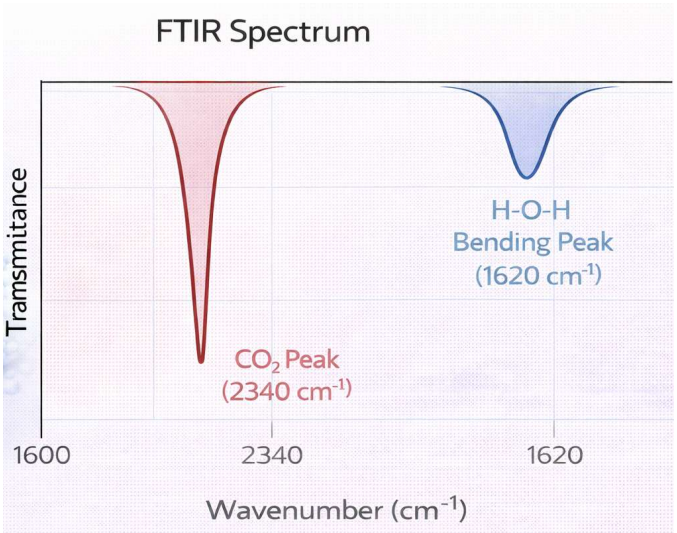
		higher vibration frequency	withdrawing substituents
Hydrogen Bonding	Reduced or disrupted hydrogen bonding	Hydrogen bonding weakens C=O bond (lower frequency); its removal increases frequency	Suggests decrease in intermolecular interactions
Steric Effects	Change in molecular environment	Steric hindrance reduces conjugation or intermolecular interactions	Leads to increase in C=O stretching frequency
Conjugation Effect	Possible reduction in conjugation	Conjugation lowers C=O frequency; loss of conjugation shifts peak higher	Indicates less resonance stabilization
Polymer Environment	Chemical modification alters local environment	Changes in polarity or neighboring groups affect vibrational energy	Confirms structural modification of polymer chain
Overall Interpretation	Shift to higher wavenumber	Combined effect of EWG, reduced H-bonding, and structural changes	Confirms successful chemical modification of polymer

6. A broad O–H stretching band is observed between 3200–3500 cm^{-1} in the FTIR spectrum of a nanocomposite. Interpret what this broadening reveals about hydrogen bonding interactions within the system.



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
O–H Peak Shape	Broad band observed between 3200–3500 cm^{-1}	Broadening occurs due to multiple O–H environments with different bonding strengths	Indicates non-uniform hydrogen bonding
Hydrogen Bonding	Strong and extensive hydrogen bonding present	Hydrogen bonding weakens O–H bond → lowers and spreads vibration frequencies	Confirms intermolecular interactions in system
Peak Width	Wide range of absorption instead of sharp peak	Distribution of hydrogen bond strengths leads to range of vibrational energies	Indicates heterogeneous structure
Molecular Interaction	Interaction between polymer and nanofiller	Functional groups (–OH) interact via H-bonding with neighboring molecules or surfaces	Suggests good interfacial interaction
Structural Environment	Different local environments for O–H groups	Variations in polarity and bonding surroundings affect IR absorption	Indicates complex nanocomposite structure
Peak Position	Slight shift toward lower wavenumber (often observed)	Hydrogen bonding reduces bond strength → lower frequency absorption	Confirms presence of strong hydrogen bonding
Overall Interpretation	Broad O–H band instead of sharp peak	Combined effect of strong and varied hydrogen bonding interactions	Confirms extensive hydrogen bonding in nanocomposite system

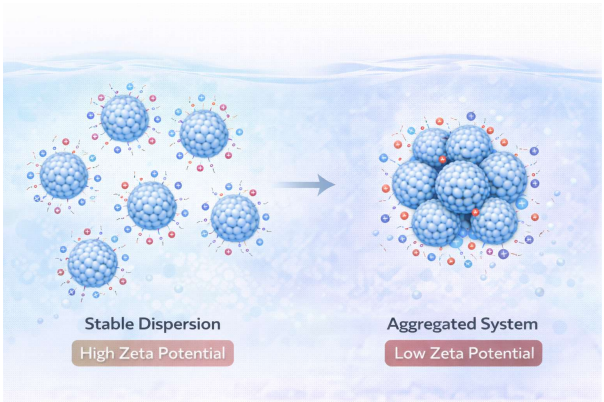
7. **TiO_2 nanopaticles exposed to atmospheric CO_2 show new peaks at 2340 cm^{-1} (CO_2) and 1620 cm^{-1} (H–O–H bending). Discuss what these peaks reveal about surface adsorption and environmental effects.**



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
CO_2 Peak	New peak appears at 2340 cm^{-1}	CO_2 molecules adsorb on TiO_2 surface, showing characteristic asymmetric stretching vibration	Confirms CO_2 adsorption on nanoparticle surface
H–O–H Bending Peak	Peak observed at 1620 cm^{-1}	Adsorbed water molecules (moisture) exhibit H–O–H bending vibration	Indicates presence of surface-adsorbed water
Surface Adsorption	Appearance of new peaks after exposure	High surface area of nanoparticles promotes adsorption of atmospheric gases	Demonstrates surface sensitivity of nanomaterials
Environmental Interaction	Peaks arise due to exposure to air (CO_2 + H_2O)	Nanoparticles interact with surrounding environment through physisorption or chemisorption	Confirms environmental influence on material properties

Surface Chemistry	Change in FTIR spectrum compared to fresh sample	Adsorbed species modify surface functional groups and bonding	Indicates alteration in surface chemistry
Overall Interpretation	New peaks not present in original sample	Combined adsorption of CO_2 and moisture from atmosphere	Confirms strong environmental adsorption effects on TiO_2 nanoparticles

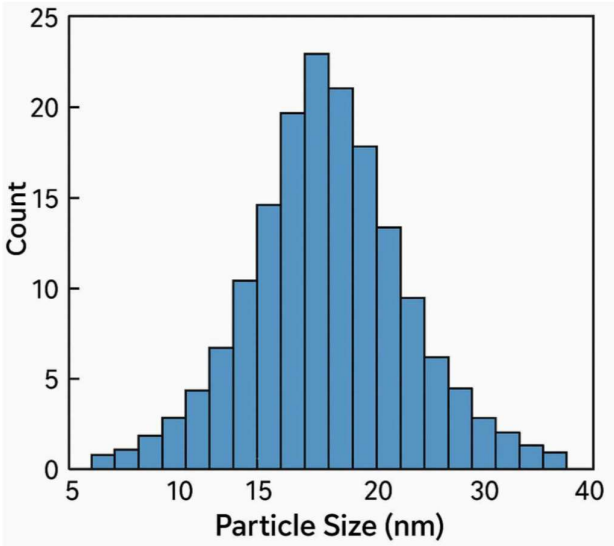
8. Differentiate between: Stable dispersion (high zeta potential) Aggregated system (low zeta potential)



Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Zeta Potential Value	High in stable system; low in aggregated system	High surface charge creates strong electrostatic repulsion; low charge reduces repulsion	High zeta potential → **stable**, low → **unstable**
Particle Distribution	Stable: well-dispersed particles; Aggregated: clustered particles	Strong repulsion keeps particles apart; weak repulsion allows attraction	Determines **dispersion vs aggregation**

Interparticle Forces	Stable: repulsive forces dominate; Aggregated: attractive forces dominate	Balance between **electrostatic repulsion and van der Waals attraction**	Governs **colloidal behavior**
Electrical Double Layer	Thick and strong in stable system; compressed in unstable system	High charge forms a strong double layer; low charge weakens it	Affects **particle interaction distance**
System Stability	Stable system shows no aggregation ; unstable system shows aggregation	Reduced surface charge leads to particle collision and sticking	Indicates **colloidal stability level**
Particle Size (Effective)	Stable: smaller effective size; Aggregated: larger clusters	Aggregation increases hydrodynamic size	Confirms **cluster formation**
Overall Interpretation	Clear difference between dispersed and clustered systems	Zeta potential controls interaction forces between particles	Determines **stability of nanoparticle suspension**

10 . A TEM histogram shows a broad particle size distribution ranging from 5–40 nm. Explain what this suggests about nucleation and growth uniformity.



A TEM histogram shows a broad particle size distribution ranging from 5-40 nm

Aspect	Observation	Reason (Scientific Explanation)	Conclusion
Particle Size Distribution	Wide range from 5–40 nm	Particles formed at different times and conditions	Indicates polydispersed system
Nucleation Process	Not uniform	Continuous or multiple nucleation events occurred instead of burst nucleation	Suggests non-uniform nucleation
Growth Uniformity	Uneven growth of particles	Different particles experience different growth rates due to local variations (concentratio	Indicates non-uniform growth

		n, temperature, surfactant coverage)	
Size Variation	Presence of both small and large particles	Early nuclei grow larger while later-formed nuclei remain smaller	Confirms time-dependent growth process
Reaction Control	Poor control over synthesis conditions	Lack of controlled kinetics leads to variability in nucleation and growth	Indicates uncontrolled synthesis process
Polydispersity	High size dispersion	Broad histogram reflects wide distribution of sizes	Confirms high polydispersity
Overall Interpretation	Broad TEM histogram	Combined effect of non-uniform nucleation and variable growth rates	Results in heterogeneous nanoparticle system